

Team 27

Quality Attribute Scenarios

Eduardo Lopez (Testability):

Stimulus: The developers will test for any syntax and logical errors. Testing against user inputs like if they click the wrong section or button a mini game or learn/game mode menu. I/O and out of memory testing involving memory issues like if the user/dev goes beyond a desired memory range. Testing will also be done on possible hardware scenarios like if the computer shut off during a mini game.

Stimulus Source: The developers with the help of some tools (e.g Unit test, CI,)

Response: The testing can be done through different path scenarios like if a certain input leads to the game mode -> mini game -> back to learn mode -> back to the mini game. Capturing how and why the I/O or out of memory error occurred. Tracking and monitoring the current frame, mini game, level of the current state of the game in which the error occurred.

Response Measure: We want to measure the time it takes to perform the tests provided by Oracle's JUnit system. We also want to measure the rate of success on our JUnit tests and how accurately they depict our systems defects.

Environment: All the tests above will be done at every sub section of the code, at every commit and push to github, and the entire system against memory issues and hardware issues. Prototyping with various users.

Artifact: The developers will run various tests from single lines of code to sub sections. Components of the frames like the level navigation, popups like incorrect/correct responses in game mode, mini game input like clicks, etc., and finally the entire system as whole will be tested.

Iris Chou (Performance)

Stimulus: Pressing of a key on the player's keyboard

Stimulus Source: A player

Response: The player's character should be redisplayed with an adjusted position to reflect the movement input from the stimulus. The new position of the character should be evaluated based on the minigame end conditions, and the minigame's subsequent screen should be displayed if those conditions are met.

Response Measure: Latency between stimulus and the on-screen display; the time between when the i/o process first logs the keyboard input and when the browser display is rerendered should be <100ms

Environment: Normal operation in play mode with a minigame actively displayed on screen

Artifact: Frontend and backend components of the web application

Eric Zhao (Usability)

Stimulus: The end user wants: ease of learning, efficient usage of students' time to learn the material, and have it be easily accessible for educational officials to add course material.

Stimulus Source: The end user (in our case, a high school student) is our primary source of stimulus for usability. Educational officials might also be a source of stimulus as curriculum is decided by CollegeBoard officials (in our case) and teachers [On a side note, the curriculum developer portion may entirely be optional or omitted for the sake of time and difficulty].

Response: End users should be given a straightforward, chronological, playthrough of historical events. It should also anticipate both students' needs and CollegeBoard needs as well as provide appropriate feedback to student performance in the game.

Response Measure: We should be calculating how long and the number of attempts it takes for the average student to perform well, the time it takes to answer a question correctly, the ratio of time a player spends in the Learn mode, the number of errors, and the ratio of wins to the total number of attempts. As for the administrator part, we should calculate the average time it takes for a teacher or official to create an entire section, the amount of time a teacher is spending learning the developer tools and actually creating the game, and the number of errors

Environment: Testing should be done in the earlier phases of level prototyping as users may have concerns about data collection mid-game. The application and developer tools are also easily downloadable and accessible on school machines.

Artifact: Players should be given a simple point and click GUI. It should also support touch screens. The developer tools should have a simple and straightforward GUI.

Joanna Acosta Rico - (Availability)

- **Stimulus:** The game is running, but the device experiences an unexpected shutdown or restart.
- **Stimulus Source:** The operating system or user triggers a sudden shutdown or restarts their device, which interrupts the game while it's running.

- **Response:** The game should allow the player to resume from their last known progress, minimizing the impact of the unexpected shutdown.
- **Response Measure:** The game should automatically save the player's progress after each level or minigame, so that the data isn't lost in case of a shutdown. When the game is launched after an unexpected shutdown, the player should be allowed to continue from where it was last saved.
- **Environment:** Game is running on a local device when an unexpected shutdown happens.
- **Artifact:** The game and the player's progress information like what level they are on and how many points they've earned

Javy Aguinaldo (Modifiability)

Stimulus: Dev implementing a new level

Stimulus Source: Designer or Dev

Response: The designer can add or update a level by editing the metadata and adding assets, thus making the pipeline to validate and create the level available.

Response Measure: A dev should complete validation and verification of a new level within 2 hours, and development itself should take less than 7 hours.

Environment: Local machine within the project tooling interface, levels stored in bundles within repos.

Artifact: Level data, authoring tools, and building processes.

Krishna Rastogi (Deployability)

Stimulus: A new version of the web application with a new UI and more levels and content is ready for deployment and the current version needs to be rolled back.

Stimulus Source: We as the developers want to deploy the newest version of the application and also roll back the previous version OR users request new updates to the game

Response: Developers initiate deployment and the new version must pass all the necessary checks and tests before replacing the active version.

Response Measure: We should track the total deployment time, downtime of the system, and the success rate of deployment. We want to have no more than 5 minutes of downtime and maintenance during deployment. Also we should track the number of changes that were made between versions.

Environment: The game is being hosted on a web server and has active users. The game must still be playable during deployment of new versions.

Artifact: The apps user interface, level data and content, or backend updates.